What Is a Filesystem?

A filesystem is a software construct to manage the storage in a partition, regardless if the partition is contained in a virtual disk, physical disk, or Logical Volume (according to the Ubuntu wiki, logical volumes correspond to partitions: they hold a filesystem). A Linux system reads and writes files unlike other systems that are "disk sector"-oriented. The read or writes initiated by the user or user program are sent to the VFS (Virtual Filesystem) layer on the way to the device driver. The advantage to the VFS layer is the user environment sees all filesystems the same, a device to read and write, while the filesystem organizes the disk and maintains the appropriate control structures.

The filesystem does all the hard work, not the user. In general terms, the filesystems have control blocks that are associated with files. The control blocks keep the file metadata, such as filename, permissions, owner, access time, creation time, and more. These control blocks are commonly known as index nodes or, simply, *inodes*. The inodes also point to the data blocks used by the file. There is a finite amount of space in an inode, so there may be intermediate pointer blocks to allow for more blocks; this is known as “indirect block pointers”. Large numbers of indirect blocks or levels of indirection may be used.

This type of filesystem is a little fragile if the system were to crash, as many of the filesystem components are cached in memory. An option that most filesystems have is what is commonly known as *journaling*. This filesystem level function writes information to a “journal” first, then writes the data to the correct location in the filesystem, and if that process is complete, the journal is released and the next operation is started. Journaling of the metadata is highly recommended in almost every environment.

You can see an example of data blocks in an inode in the image below: